VIRANGA LAKSHAN

SOFTWARE DEVELOPER

Medical Technology Engineering • Web Applications, APIs, Cloud & AI Systems

Target Company: Wosler Diagnostics Inc.
Role: Software Developer (Reporting to CSO)
Arrangement: Virtual / Full-Time
Availability: Immediate Start

Engineering Mission & Alignment with Wosler

Full-Stack Software Engineer with **5+ years of production experience** building resilient web applications, REST APIs, relational database backends, and cloud-hosted healthcare systems. Deeply aligned with Wosler's mission to expand diagnostic imaging access across Alberta, Saskatchewan, and Nova Scotia. Experienced delivering clean, maintainable, and audit-ready software within production environments supporting real-world clinical and patient workflows.

5+ Years	React & TS	REST & SQL	AWS & AI
FULL-STACK SOFTWARE ENG.	MODERN WEB APPLICATIONS	APIS & RELATIONAL DATA	CLOUD, AUTOMATION & LLMS

1. QUALIFICATIONS & REQUIREMENTS ALIGNMENT MATRIX

Direct Mapping to Wosler's Technical & Operational Criteria

WOSLER ROLE REQUIREMENT	VIRANGA'S PRODUCTION EXPERIENCE & ENGINEERING COMPETENCY	STATUS & MATCH
3+ Years Software Development	5+ years building and maintaining enterprise software in production, contributing across frontend UI, backend services, and automated CI/CD pipelines.	**Exceeds (5+ Yrs)**
Modern Frontend (React, Vue)	Extensive production engineering in React and TypeScript; building modular component architectures, accessible patient portals, and clinician dashboards.	**Core Competency**
Backend & REST APIs	Architecting scalable RESTful web APIs with Node.js/TypeScript, Python, and C#; integrating microservices, third-party medical devices, and external webhooks.	**Production Tested**
Relational Databases & SQL	Designing normalized schemas, indexing hot paths, tuning execution plans, and enforcing strict ACID transaction boundaries in PostgreSQL and SQL Server.	**Advanced SQL**
Clean Code, Testing & Git	Disciplined practitioner of clean architecture, SOLID principles, TDD, automated unit/integration testing (Jest, PyTest), Git pull requests, and peer reviews.	**Standard Practice**
Security & Sensitive Data	Deep understanding of application security: OAuth2/JWT auth, Role-Based Access Control (RBAC), TLS 1.3 encryption, and compliance principles (PIPEDA / HIPAA).	**Compliance Ready**
Cloud, Docker & Linux (Nice-to-Have)	Hands-on AWS experience (S3 for imaging files, Lambda serverless tasks, EC2, RDS), Docker containerization, and automated GitHub Actions CI/CD workflows.	**AWS & Containers**
AI/ML & Automation (Nice-to-Have)	Integrating LLM APIs and machine learning models for clinical note extraction, report summarization, automated requisition triage, and workflow routing.	**Hands-On AI/LLM**

2. ENGAGEMENT OVERVIEW & READINESS

Immediate Full-Time Availability Aligned with Wosler's Growth

Full-Time Commitment — CAPACITY

Dedicated full-time capacity reporting to the CSO, integrating seamlessly with Wosler's internal engineering team and technical partners.

Compensation Model — BASE SALARY

Fully aligned with Wosler's policy: open and flexible to discuss standard competitive base salary ranges during the first introductory conversation.

Immediate Availability — DAY 1 READY

Ready to start immediately within 24–48 hours; prepared to set up development environments, review documentation, and pick up initial sprint tasks.

FULL-STACK ARCHITECTURE FRONTEND & BACKEND

React, TypeScript, RESTful Services, SQL Architecture & Healthcare Security

Frontend: React • TypeScript • State Management • Responsive UI
Backend: Node.js • Python • REST APIs • Microservices
Database: PostgreSQL • SQL Server • ACID Transactions • Indexing

3. MODERN FRONTEND ENGINEERING (REACT, TYPESCRIPT & CLINICAL UX)

Intuitive, High-Reliability Interfaces for Healthcare Teams and Patients

React & TypeScript Component Architecture — FRONTEND CORE

Building maintainable, type-safe web applications for clinical environments:

- **Strict Type Safety:** Leveraging TypeScript interfaces to eliminate runtime errors across complex clinical data models (demographics, study metadata, doctor orders).
- **Modular Reusable Design System:** Building accessible, responsive component libraries ensuring consistent UX across reception desks, sonographer tablets, and patient portals.
- **Optimized State Management:** Implementing React Query / Zustand for efficient server-state caching, eliminating redundant network calls during busy clinic shifts.
- **Error Boundaries & Resilient UX:** Designing graceful degradation so clinical staff never experience white-screen crashes during active patient consultations.

Clinic Operations & Patient-Facing Applications — CLINICAL WORKFLOWS

Engineering software tailored for fast-paced clinic operations:

- **Patient Intake & Check-in:** Self-service digital check-in flows capturing health cards, requisition uploads, and medical history with clear validation.
- **Clinic Scheduling & Queue Boards:** Real-time room and modality status views (Ultrasound, X-ray) that minimize patient wait times and track technologist utilization.
- **Requisition & Document Viewers:** Rapid document rendering with zooming, annotation, and metadata extraction to assist clinical coordinators.
- **Offline-Tolerant Patterns:** Optimistic client updates with background queue retries for clinics operating in remote areas with intermittent connectivity.

4. BACKEND SERVICES, REST APIS & RELATIONAL DATABASE DESIGN

High-Throughput, ACID-Guaranteed Data Management

RESTful API Design — SERVICES

Designing clean, versioned REST endpoints with OpenAPI/Swagger contracts, predictable JSON schemas, semantic HTTP status codes, and rate limiting.

Relational SQL Schemas — INTEGRITY

Architecting normalized schemas in PostgreSQL and SQL Server with foreign key constraints, composite B-tree indexes, and optimized query plans.

ACID Data Guarantees — RELIABILITY

Enforcing strict database transaction boundaries and idempotent write endpoints, guaranteeing zero duplicate orders and zero dropped patient records.

5. APPLICATION SECURITY & HEALTHCARE DATA PRIVACY (PIPEDA / HIPAA STANDARDS)

Protecting Sensitive Patient Health Information (PHI) Across Canadian Jurisdictions

Authentication & Access Control — IDENTITY

- **Role-Based Access Control (RBAC):** Enforcing granular permission tiers separating receptionists, sonographers, radiologist readers, and clinic managers.
- **OAuth2 & JWT Token Architecture:** Secure token generation with short TTLs, rotating refresh tokens, and CSRF/XSS protection.
- **Multi-Factor Authentication (MFA):** Mandating MFA protocols for administrative and clinical access points to meet healthcare audit standards.

Data Protection & Audit Traceability — PRIVACY COMPLIANCE

- **Encryption Standards:** Enforcing TLS 1.3 for data in transit and AES-256 for data at rest across database tables and imaging file storage.
- **Immutable Audit Trails:** Logging all clinical chart views, export actions, and modifications to satisfy provincial privacy regulations (e.g., Alberta HIA).
- **Data Sanitization & De-Identification:** Masking patient health card numbers and personal identifiers in test fixtures and logging outputs.

Software Engineering Hygiene & Quality Assurance

High code standards are non-negotiable in healthcare technology. I build automated unit test suites (Jest/PyTest) for business validation rules, integration tests for API contracts, and perform thorough manual smoke testing before submitting clean, well-documented pull requests.

CLOUD & MEDTECH AWS, AI & STANDARDS

AWS Cloud Architecture, Docker, AI Integration & Diagnostic Imaging Protocols

Cloud: AWS (S3, Lambda, EC2, RDS) • Docker • Linux
MedTech Protocols: DICOM • PACS/RIS • HL7 / FHIR Awareness
AI Integration: LLM APIs • Structured Parsing • Audio Transcription

6. AWS CLOUD INFRASTRUCTURE, CONTAINERIZATION & CI/CD

Scalable, Resilient Hosting for Distributed Clinical Networks

AWS Healthcare Cloud Services — AWS STACK

Hands-on experience deploying and managing cloud workloads on AWS:

- **Amazon S3 for Medical Storage:** High-throughput, HIPAA/PIPEDA-compliant storage for ultrasound/X-ray studies, using pre-signed URLs and lifecycle archiving.
- **AWS Lambda Serverless Workers:** Triggering event-driven background tasks (DICOM metadata indexing, automated report notifications, batch data sync).
- **Amazon RDS (PostgreSQL):** Multi-AZ high-availability database instances with automated encrypted daily backups and point-in-time recovery.
- **AWS IAM Security:** Principle of least privilege with strictly scoped role-based access for services and developers.

Docker, Linux & CI/CD Deployment Pipelines — DEVOPS AUTOMATION

Building repeatable, containerized deployment pipelines:

- **Docker Containerization:** Packaging frontend apps and backend microservices into lightweight, reproducible Alpine/Debian Linux containers.
- **GitHub Actions CI/CD:** Automated pipelines running linters, TypeScript type checks, unit tests, and security dependency scans on every commit.
- **Production Rollouts:** Staged release environments (Dev → Staging → Production) enabling zero-downtime rolling updates across clinic endpoints.
- **Environment Configuration:** Managing secrets and environment variables cleanly through secure key management (AWS Secrets Manager / SSM).

7. ISO 13485 & DIAGNOSTIC IMAGING INTEROPERABILITY AWARENESS

Engineering within Regulated Medical Device & Quality Management Frameworks

ISO 13485 & MDSAP — COMPLIANCE

Familiar with the rigorous documentation, traceability (IEC 62304), and verification standards required for certified medical device software operating in clinical clinics.

DICOM & PACS/RIS — IMAGING FLOW

Strong architectural understanding of diagnostic imaging pipelines: DICOM study metadata, C-STORE transfers, PACS image archiving, and RIS scheduling integration.

HL7 & FHIR Standards — INTEROPERABILITY

Grounded in modern health data exchange: mapping clinical events to FHIR resources (`Patient`, `ImagingStudy`, `DiagnosticReport`) for hospital interoperability.

8. AI-POWERED HEALTHCARE TOOLS, LLMS & WORKFLOW AUTOMATION

Practical AI Solutions That Streamline Clinical Operations and Reduce Provider Fatigue

Intelligent Requisition & Triage Automation — DOCUMENT AI

Integrating LLMs and optical character recognition (OCR) to parse incoming doctor referral PDFs, automatically extracting patient demographics, requested examination protocols, and urgency flags into structured database records to accelerate scheduling.

Medical Audio Transcription & Structured Reporting — VOICE & REPORTING

Connecting speech-to-text models (e.g., fine-tuned Whisper) to transcribe radiologist and sonographer audio findings in real time, automatically populating standardized diagnostic report templates with human-in-the-loop review safeguards.

The Seed-Stage Engineering Mindset

As Wosler grows post-seed, software development requires both agility and discipline. I bring an entrepreneurial, curious problem-solving attitude: comfortable tackling undefined problems, working across the entire technology stack, and taking pride in getting working, dependable software into the hands of clinical teams.

DELIVERY & GOVERNANCE

CASE STUDIES & PLAN

Engineering Track Record, Project Management Support & 30-Day Value Roadmap

Software Engineering: Viranga Lakshan (Full-Stack Engineer)
Project Management: Lithumitha Ravikumar (Project Manager)
Readiness: Immediate Start • Full-Time Remote

9. TEAM ALLOCATION & PROFESSIONAL ROLES

Clear Engineering Execution Backed by Dedicated Project Management

Viranga Lakshan — FULL-STACK SOFTWARE DEVELOPER

Primary Software Engineering Lead

- Personally responsible for full-stack software development across React, TypeScript, APIs, and databases.
- Designs, tests, and deploys clinical tools, workflow automations, and backend services.
- Troubleshoots production issues, investigates performance bottlenecks, and maintains automated test suites.
- Collaborates directly with the CSO, internal engineering teams, and contracted development partners.

Lithumitha Ravikumar — PROJECT MANAGER

Dedicated Project Management Support

- Coordinates sprint schedules, requirements traceability, and delivery milestones.
- Manages documentation quality assurance to support ISO 13485 and MDSAP compliance workflows.
- Maintains cross-functional communication, issue tracking, and sprint status reporting.
- *Note: Role is dedicated project management and delivery coordination.*

10. RELEVANT PRODUCTION CASE STUDIES

Demonstrated Technical Excellence in Mission-Critical, Distributed Systems

High-Throughput Mobile & Clinic Network (MFCS)

400+ SITES • 1M+ DAILY WRITES

Architecture & Scope: Distributed transaction and record collection platform across 400+ nationwide retail branch sites:

- **0.00% Record Loss:** Strict ACID transaction boundaries processing 1,000,000+ daily operational records with zero data loss.
- **Zero Database Freezes:** Eliminated row-level lock contention, reducing lock wait times from 1,450 ms to 45 ms.
- **Cloud Modernization:** Migrated database workloads to AWS RDS with continuous production uptime.

Healthcare Portfolio & Analytics Platform

SECURITY & ANALYTICS

Architecture & Scope: Secure, multi-tenant analytics and reporting portal for institutional stakeholders:

- **React & TypeScript UI:** Interactive, high-density data visualizations and role-gated reporting dashboards.
- **Audit & Security Compliance:** Role-based access control (RBAC), multi-tenant data isolation, and encrypted audit trails.
- **Automated Testing:** Comprehensive Jest and Cypress test coverage ensuring zero-regression releases.

11. RAPID 30-DAY ONBOARDING & VALUE DELIVERY ROADMAP

Structured Transition to Accelerate Wosler's Clinical Software Velocity

Days 1 – 10: Ramp-Up — FOUNDATION

Set up local dev environment, review architecture and ISO 13485 guidelines, shadow clinic workflows, and ship first bug fix / UI enhancement PR.

Days 11 – 20: Feature Velocity — EXECUTION

Take ownership of full-stack sprint tickets (React, REST APIs, SQL), write automated tests, participate in code reviews, and optimize clinic tools.

Days 21 – 30: Autonomous Drive — OWNERSHIP

Lead end-to-end feature implementations independently, contribute to AI/automation initiatives, and actively support production deployments.

Ready for Initial Technical Conversation & Interview

I am available immediately for an introductory conversation with the CSO and engineering leadership to discuss Wosler's software roadmap, review code samples, and explore how my full-stack skills can support your clinic network.